

claude-windows / 6a82308f-69d6-43a8-9ea4-d8eaf497253d

Metadata

- Source: `claude-windows`
- Kind: `claude`
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\6a82308f-69d6-43a8-9ea4-d8eaf497253d\subagents\workflows\wf\_736be611-fb5\agent-a128ad3b22e13bee4.jsonl`
- SHA-256: `3485a996e0409aee221fe4dd94cf2602df666687310731f8e75db07689a9d16a`
- Source modified: `2026-06-16T15:16:32+00:00`
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- project: `wf\_736be611-fb5`
- session_id: `6a82308f-69d6-43a8-9ea4-d8eaf497253d`

Transcript

1. user (2026-06-16T15:14:30.302Z)

Adversarially VERIFY this finding from the real-footage rally-detection validation. Try hard to REFUTE or show it is overstated, using ONLY the data (you MAY Read output/local_vs_gemini_6.json). Default to holds=false if the numbers do not clearly support it.

CLAIM (trajectory-signal): mahadevpura-2 'fits' because its inter-rally dead time produces clean inactivity gaps at roughly true-rally cadence, so the cap splits at real rally boundaries.
EVIDENCE CITED: mahadevpura-2 W1: n=40 vs Gemini 39, count_ratio 1.03, mean 13.8s, max 43.3s; on the owner's 40-label ground truth the hard-cap variant reaches F1@0.25 0.722 and mean-IoU 0.50 ? by far the best local fit, vs baseline 0.000/0.03.

REAL-FOOTAGE VALIDATION DATA (6 Kushagra/Yash June-12 clips, 1080p proxies, fully-LOCAL no-AI WASB windowing vs Gemini. Gemini = strong REFERENCE, NOT ground truth ? these 6 have no golden labels EXCEPT mahadevpura-2 which the owner hand-labeled, see GROUND-TRUTH below).

Rally COUNT (Gemini | local baseline | local W1 cap=12 | local W1+W2 mc=1):
GX010128: 39 | 15 | 78 | 75
GX010129: 27 | 2 | 50 | 47
GX020125: 11 | 6 | 18 | 15
mahadevpura-0: 40 | 19 | 62 | 43
mahadevpura-1: 9 | 1 | 2 | 2
mahadevpura-2: 39 | 5 | 40 | 40
AGGREGATE: Gemini 165 | baseline 48 (mean window 62.9s, 27 merge-groups, mIoU 0.23) |
W1 250 (mean 10.8s, 10 merge-groups, mIoU 0.39) |
W1+W2 222 (mean 11.5s, 10 merge-groups, mIoU 0.39, 6 gemini-only "misses")
Per-video mean window (W1): GX010128 7.8s, GX010129 9.3s, GX020125 14.1s, mahadevpura-0 10.4s,
mahadevpura-1 86.9s (the continuously-active blob W1 cannot cap ? no inactivity gap),
mahadevpura-2 13.8s. Gemini means ~5.7-7.2s.

GROUND-TRUTH (mahadevpura-2, 40 owner-labeled rallies, mean 5.7s, verified frame-aligned):
F1@0.5 / F1@0.25 / mean-IoU: Gemini 0.557/0.835/0.55 ; local W1+W2 0.150/0.500/0.31 ;
local W1+W2+hard-cap 0.296/0.722/0.50 (best local). baseline 0.000/0.000/0.03.
BOUNDARY ERROR (matched rallies, seconds): Gemini [?start| med 1.46 |?end| med 1.19 ;
local-hardcap [?start| med 1.74 (~= Gemini!) |?end| med 3.12 (> Gemini).
=> local's rally-START detection is ~at par with Gemini; the gap is the rally-END.
IoU<->time: for a 5.7s rally, tIoU 0.5 tolerates only ~+-1.9s shift (the ~1-1.5s service window
nearly exhausts it -> tIoU 0.5 over-penalizes; owner wants a +-2s tolerance metric).

PRIOR GOLDEN DELTAS (n=6, then n=7 incl. mahadevpura-2): baseline F1 0.300 -> W1 0.439 (+0.139, 6/0)
-> W1+W2 0.457 (+0.019, 6/0); hard-cap golden-neutral on n=6 (+0.006 n.s.) but became BEST on

n=7

(continuously-active clips). All flags default-OFF, promoted only on measured LOVO lift + sign-test.

WINDOWING KNOBS: a frame is "active" iff shuttle visible AND per-second-normalized velocity > velocity_thresh(0.02); active frames within merge_gap(2.0s) coalesce; min_window(2.0s). W1 = max_window_duration cap(12s) split at the largest internal inactivity gap. W1.5 hard-cap = midpoint-chop when no gap. W2 = net-crossings gate (drop windows with <1 net crossing).

2. assistant (2026-06-16T15:14:33.620Z)

I'll verify this claim by examining the actual data file and checking the cited numbers against it.

3. user (2026-06-16T15:14:34.459Z)

File does not exist. Note: your current working directory is C:\Users\avidu\Projects\badminton-highlight-indexer\.claude\worktrees\w2-safeguard.

4. assistant (2026-06-16T15:14:37.071Z)

The file isn't at the expected path. Let me locate it.

5. user (2026-06-16T15:14:37.511Z)

No files found

6. user (2026-06-16T15:14:40.040Z)

eval_baselines\gemini_wrapper_2026-06-10.json

7. assistant (2026-06-16T15:14:44.598Z)

The cited data file `output/local\_vs\_gemini\_6.json` does not exist in this worktree. Let me search more broadly for any file containing the relevant data (mahadevpura-2, etc.) so I can verify against actual numbers rather than just the prompt's restated figures.

8. user (2026-06-16T15:14:45.459Z)

Found 6 files

docs\RALLY_QUALITY_RESEARCH.md

docs\QUALITY_ITERATIONS.md

backend\pipeline\segmenters\fusion_hybrid.py

eval_baselines\gemini_wrapper_2026-06-10.json

eval_baselines\heuristic_lovo_n6.json

docs\archives\quality-iterations\2026-06-first-ab-experiment\README.md

9. user (2026-06-16T15:14:48.613Z)

C:\Users\avidu\Projects\badminton-highlight-indexer\.claude\worktrees\w2-safeguard\docs\RALLY_QUALITY_RESEARCH.md-72-

C:\Users\avidu\Projects\badminton-highlight-indexer\.claude\worktrees\w2-safeguard\docs\RALLY_QUALITY_RESEARCH.md-73-### 2.1 L1 (shuttle tracking) is NOT the bottleneck

C:\Users\avidu\Projects\badminton-highlight-indexer\.claude\worktrees\w2-safeguard\docs\RALLY_QUALITY_RESEARCH.md-74-On the divergence clips WASB reported the shuttle visible in **96.99 % of frames** (mahadevpura-1

docs\RALLY_QUALITY_RESEARCH.md:75:98.6 %, GX020125 72.8 %, mahadevpura-2 78.8 %). The trajectory signal is present and dense.

C:\Users\avidu\Projects\badminton-highlight-indexer\.claude\worktrees\w2-safeguard\docs\RALLY_QUALITY_RESEARCH.md-76-

C:\Users\avidu\Projects\badminton-highlight-indexer\.claude\worktrees\w2-safeguard\docs\RALLY_QUALITY_RESEARCH.md-77-### 2.2 L2

(segmentation) is the bottleneck ? and the flaw is conceptual
C:\Users\avidu\Projects\badminton-highlight-indexer\claude\worktrees\w2-safeguard\docs\RALLY_QUALITY_RESEARCH.md-78-The windowing is
`trajectory\_to\_action\_windows`

10. assistant (2026-06-16T15:14:51.273Z)

Let me read the relevant sections of the research docs to find the actual recorded numbers for mahadevpura-2, the W1 cap, and the hard-cap variant.

11. user (2026-06-16T15:14:52.490Z)

```
1      # Rally-Detection Quality: Research, Eval Framework & Project Plan
2
3      **Status:** Foundation doc (2026-06-16), run as a **tracked project** (§7) ? **Bridge
the
4      local?Gemini rally-detection gap,"* **completed when all committed tiers land and a
single quality
5      number is attributed (§7.1).** The empirical + research backbone for closing the gap
between the
6      **local, no-cloud rally segmenter** and the **Gemini** path. Iterations land as their
own measured
7      PRs, logged in [`QUALITY_ITERATIONS.md`](QUALITY_ITERATIONS.md); this doc is the plan
they execute against.
8
9      **How to read this:** §1 frames the gap with measured numbers. §2 is the precise, code-
grounded
10     diagnosis (read before proposing fixes). §3 inventories the eval/experiment harness we
already
11     have. §4 specifies how we strengthen that harness (the "really strong evaluation
framework").
12     §5 is the cited external-research synthesis (4 dimensions). §6 is the prioritized
roadmap with
13     falsifiable success criteria. **§7 is the project plan ? work-item tracker, sequencing,
the single
14     number we attribute to the effort, and the definition of done.** §8 recaps the non-
negotiable
15     constraints + cross-links.
16
17     > **This doc does NOT duplicate prior design.** It builds on, and cross-links:
18     > [`HOW_RALLY_DETECTION_WORKS.md`](HOW_RALLY_DETECTION_WORKS.md) (the 2-layer mental
model),
19     > [`EXPERIMENT_HARNESS.md`](EXPERIMENT_HARNESS.md) (the A/B + LOVO harness, P1 shipped),
20     > [`MULTI_SIGNAL_FUSION_PLAN.md`](MULTI_SIGNAL_FUSION_PLAN.md) (shuttle+person+audio
fusion),
21     > [`ANALYZERS_AND_RECONCILERS.md`](ANALYZERS_AND_RECONCILERS.md) (the signal-fusion
framework +
22     > §13 literature review ? **already** establishes that our skeleton is standard TAL/TAS
+
23     > late-fusion; we *cite, don't re-derive*), [`ABLATION_LAYER.md`](ABLATION_LAYER.md),
24     > [`EVALUATION.md`](EVALUATION.md),
[`OWNED_MODEL_IMPLEMENTATION_PLAN.md`](OWNED_MODEL_IMPLEMENTATION_PLAN.md)
25     > (TAL-head-first ? VLM), and [`GOLDEN_SET_VENDORING.md`](GOLDEN_SET_VENDORING.md)
(corpus growth).
26
27     ---
28
29     ## 1. The gap (measured)
30
31     Two paths produce rally segments today. **Gemini** (cloud VLM, watches the video
semantically)
32     is strong; the **local** path (WASB shuttle trajectory ? velocity windowing, no cloud)
lags
33     badly. Measured F1 vs golden labels ([`QUALITY_ITERATIONS.md`](QUALITY_ITERATIONS.md)):
34
```

```

35 | Path | Mahadevpura (clean, single court) | Kushagra (multi-court) |
36 |---|---|---|
37 | Heuristic velocity windowing (local today) | 0.657 | **0.157** |
38 | Gen-0 owned head (LogReg on 5 trajectory features, LOVO n=6) | 0.744 | 0.561 |
39 | Two-stage WASB + Gemini refine | 0.83 | ? |
40 | **Gemini wrapper (gemini-flash)** | **0.93** | **0.66** |
41
42 The local heuristic is **footage-fragile**: fine on a clean single court, collapses on
43 multi-court / busy venues. The owned Gen-0 head already closes most of the multi-court
gap
44 (0.561 vs 0.66) ? the residual is **corpus-sized, not architecture-sized**.
45
46 ### 1.1 The 2026-06-16 local-vs-Gemini divergence run (no golden labels)
47
48 A fully-local run (`wasb_hybrid --skip-ai-handoff`) vs Gemini on 3 fresh matches
49 (Kushagra/Yash June-12; report: `~/June 12 Local-vs-Gemini/_LOCAL_vs_GEMINI_REPORT.md`,
50 generator `scratch/compare_local_vs_gemini.py`). **Gemini is a strong reference here,
NOT
51 ground truth** (these clips are unlabeled) ? so this is an agreement/divergence study:
52
53 | | Gemini | Local (WASB no-AI) |
54 |---|---|---|
55 | rallies / windows (3 videos) | 59 | 12 |
56 | total "play" | 6:28 | **17:41** |
57 | mean window duration | **6.1 s** | **65 s** (10.6x) |
58 | true misses (gemini-only) | ? | **0** |
59 | merge groups (one local window ? 2 Gemini rallies) | ? | 7 |
60
61 Headline: **the local path does not *miss* rallies ? it fails to *segment* them.** One
62 mahadevpura-1 window spanned the entire 174 s clip. Local "play" is 2.7x Gemini's
because its
63 mega-windows swallow the dead time between rallies.
64
65 ---
66
67 ## 2. Diagnosis (code-grounded ? read before proposing fixes)
68
69 The local pipeline is **two layers** (see
[HOW_RALLY_DETECTION_WORKS.md])(HOW_RALLY_DETECTION_WORKS.md)):
70 ***(L1) detection** ? per-frame shuttle position (WASB/TrackNet); and ***(L2)
segmentation** ?
71 turn the trajectory into rally intervals. The gap is almost entirely **L2**.
72
73 ### 2.1 L1 (shuttle tracking) is NOT the bottleneck
74 On the divergence clips WASB reported the shuttle visible in **96?99 % of frames**
(mahadevpura-1
75 98.6 %, GX020125 72.8 %, mahadevpura-2 78.8 %). The trajectory signal is present and
dense.
76
77 ### 2.2 L2 (segmentation) is the bottleneck ? and the flaw is conceptual
78 The windowing is `trajectory_to_action_windows`
79 ([`backend/pipeline/detectors/tracknet_runner.py:256`](../backend/pipeline/detectors/tra
cknet_runner.py)):
80
81 - A frame is **"active"** iff the shuttle is **visible** AND its **per-second-normalized
speed**
82 exceeds `velocity_threshold` (default 0.02). Velocity = `(dist / frame_width) * (fps /
dframes)`
83 ? **already fps-normalized** (? fraction of frame-width per second).
84 - Active frames within `dead_time_merge_gap` (2.0 s) are **merged** into one window.
85 - Windows shorter than `min_action_window_duration` (2.0 s) are dropped.
86
87 **The conceptual flaw: "the shuttle is moving" is treated as "a rally is in progress."**
During
88 dead time the shuttle is still visible and gently moving (being picked up, walked back,

```

knock-ups,

89 slow serve prep) ? those frames read as **active**. The 2 s merge then bridges the brief gaps, and

90 because there is ***no `max\_window\_duration` cap and no inactivity/boundary split***, a single

91 window grows until the shuttle finally goes invisible ? which, at 96?99 % visibility, is rarely.

92 Result: rally + dead-time + rally + ? collapse into a few mega-windows.

93

94 > ***Correction to an earlier hypothesis (honesty note).*** A first pass guessed the cause was

95 > "velocity threshold not fps-invariant." It IS fps-invariant (the *`fps/dframes`* factor). The

96 > real cause is the **motion-equals-rally** conflation + 2 s bridging + no upper bound. fps matters

97 > only secondarily: denser 60 fps sampling keeps more slow-motion dead-time frames above the

98 > speed threshold. The *`\_LOCAL\_vs\_GEMINI\_REPORT.md`* has been corrected to match.

99

100 ### 2.3 What this implies

101 The local path ***has no rally-STATE signal*** ? only "is the shuttle moving." Bridging the gap is

102 fundamentally about giving L2 signals that separate **rally** from **shuttle-merely-moving**

103 (net-crossings, two-sided play, sustained fast exchange, hit sounds) and a segmenter that

104 produces **bounded, well-cut** intervals ? exactly the multi-signal-fusion + learned-segmenter

105 direction the existing docs lay out. This doc makes that concrete and measurable.

106

107 ---

108

109 ## 3. What we already have (eval/experiment harness inventory)

110

111 The harness is rich and composable ? we ***extend*** it, not reinvent it. (Modules under

112 *`backend/eval/`*.)

113

114 ***Scoring & matching***

115 - *`metrics.py`* ? temporal-IoU interval matching; precision/recall/F1, mean_iou, boundary errors.

116 - *`segmentation\_metrics.py`* ? ***F1@{0.1,0.25,0.5}***, ***mAP@tIoU*** (0.3?0.7), ***segment_count_ratio***

117 (the over/under-segmentation diagnostic). (Edit/segmental score deliberately omitted to date.)

118

119 ***Statistical rigor (small-n)***

120 - *`eval\_stats.py`* ? bootstrap 95 % CIs, paired exact ***sign-test***, *`paired\_delta\_ci`*,

121 *`grouped\_folds`* (LOGO), *`out\_of\_fold\_predictions`* (anti-stacking-leakage). The promotion bar =

122 "?F1 CI excludes 0 ***AND*** sign-test agrees."

123 - *`score\_calibration.py`* ? Platt + isotonic calibrators, ***Brier + ECE***.

124

125 ***Variants, A/B, leaderboard***

126 - *`experiment.py`* ? *`Variant`* (declarative recipe: segmenter + config + cue_flags + gates +

127 learned-classifier + threshold/merge_gap/calibrate), *`compare()`* (LOVO-honest A/B with an

128 "honest" stats block), *`compare\_unlabeled()`* (agreement on unlabeled footage), *`CONTROL`*

129 (pinned baseline), *`record\_experiment()`* ? *`eval\_baselines/experiment\_leaderboard.json`*.

130 - *`ablation.py`* ? leave-one-signal-out marginal ?F1 with CI + sign-test + a no-over-claim verdict

131 vocabulary (*`earns\_keep`* / *`suggestive`* / *`inert`* / *`structural\_protected`*).

132 - *`classifier.py`* ? dependency-free numpy *`LogisticRegression`* (the learned scorer; JSON-serializable).

```

133
134     **Eval?serve discipline + gates**
135     - `windowing.py` ? single-sources the **SERVED** preset from `IndexingConfig` defaults
so eval
136         and production can't silently diverge; `PROPOSAL` (recall-oriented) is where the
golden feature
137         substrate is built. (This exists *because* Gate-A measured **+0.074 on PROPOSAL but
?0.008 on
138         SERVED** and was reverted ? the canonical eval?serve lesson.)
139     - `promotion.py` / `served_gate_a.py` ? promote a cue only if it does NOT regress at the
served
140         windowing; `per_user_gate.py` / `per_user_regression.py` ? per-user analogues;
`regression.py` ?
141         golden-set no-regression baseline with a GT-hash to detect label changes.
142
143     **Golden substrate**
144     - `gt_loader.py` ? one canonical GT loader (both legacy formats). `fusion_golden.py` /
145         `fusion_compare.py` / `fusion_audio.py` ? extract replayable per-window feature JSONs
146         (shuttle 5 + person 5 + audio 3) so ablation/compare re-score on CPU without re-
detecting.
147     - **Corpus today: 6 golden videos** (label-set hash `50d98ffbc370`); ~10 more expected
via the
148         Roora vendoring track ([`GOLDEN_SET_VENDORING.md`](GOLDEN_SET_VENDORING.md)).
149
150     **Signals available to L2 (built, default-OFF):** net-crossing count (`rally_gate.py`,
Gate-A),
151     5 person/court features (`person_detector.py` + `fusion_features.py`), 3 audio-hit
features
152     (`audio_features.py`). Fusion hooks: `fusion_hybrid.py::_enrich_candidate_stats` /
153     `_select_for_handoff`. Persisted in `candidate_segments.stats`.
154
155     ### 3.1 Known gaps in the harness (what §4 fills)
156     1. **Over-segmentation is under-surfaced in the default report.** `segment_count_ratio`
exists but
157         the merge/split *breakdown* (the very thing that exposed the over-merge) lives only
in a scratch
158         script. tIoU-F1 alone is **blind to over-merge** ? a giant window can still "match"
via overlap.
159     2. **No segmental edit-score** (Levenshtein over the predicted/gt label sequence) ? the
standard
160         TAS over-segmentation metric.
161     3. **Per-cue calibration is fitted but never *measured* post-hoc** (no ECE/Brier delta
reported).
162     4. **`out_of_fold_predictions` (anti-leakage stacking) is built but not wired to an
arbiter.**
163     5. **No active-learning loop** ? nothing tells us *which* unlabeled video to label next.
164     6. **No single "rally-segmentation eval" entrypoint** ? the pieces exist but a one-
command
165         `score this segmenter vs golden, honestly` runner would make every iteration cheap.
166     7. **The local-vs-Gemini agreement track is ad-hoc** (a scratch script), not a first-
class
167         *shadow* signal alongside the golden gate.
168
169     ---
170
171     ## 4. The strengthened evaluation framework (the "really strong harness")
172
173     Design principles (inherited, non-negotiable):
174     - **Signals ? learned scorer, NO special-casing**
([`ANALYZERS_AND_RECONCILERS.md`](ANALYZERS_AND_RECONCILERS.md) §1):
175         every new cue is a feature column the scorer learns to weight; never an `if/else` in
the decision path.
176     - **Promote only on measured lift; default OFF; revert is one flag**
([`EXPERIMENT_HARNESS.md`](EXPERIMENT_HARNESS.md) §6).
177     - **eval ? serve:** a harness win must be re-confirmed on the *served* windowing before

```

```

any default flip.
178     - Honest small-n stats: never a bare mean at  $n \geq 16$ ; always CI + paired sign-test
(C1).
179
180     4.1 Metric set the leaderboard should standardize on
181     Surface ALL of these in the default compare() report (most already computed; the task
is to
182     *report* them together so over-merge can never hide again):
183
184     - tIoU-F1 @0.5 (primary) and F1@{0.1,0.25,0.5} (segmental, TAS-standard).
185     - mAP@tIoU (0.3-0.7) when predictions carry confidences.
186     - segment_count_ratio + a merge-rate / split-rate breakdown (fraction of GT
rallies that
187     share a predicted window = under-segmentation; fraction of predicted windows that span
 $\geq 2$  GT =
188     the over-merge signal). Promote the scratch compare_local_vs_gemini.py matching into
a reusable
189     `backend/eval/` helper.
190     - Segmental edit (Levenshtein) score ? the canonical over-segmentation metric we
currently lack.
191     - Boundary timing (mean start/end error on matched pairs) ? already in metrics.py,
surface it.
192     - Calibration (ECE/Brier) reported pre/post per-cue calibration (close gap #3).
193
194     > Why: plain tIoU-F1 rewarded our mega-windows (they "overlap" real rallies). The
metric set
195     > above makes over/under-segmentation first-class, so a fix that improves segmentation
is
196     > visible even when tIoU-F1 barely moves ? and vice-versa.
197
198     4.2 Splits & significance
199     - Leave-one-video-out, grouped by match (never leave-one-window-out ? windows
within a video
200     are correlated). Keep grouped_folds. Note the RL00CV bias caveat for tiny n (see
§5.4).
201     - Bootstrap CI + paired exact sign-test on per-video held-out F1 (C1). A result is
*promotable*
202     only if CI excludes 0 and the sign-test agrees, and it survives the served-
windowing check.
203     - Anti-leakage: per-LOVO-fold threshold tuning + calibration on the TRAIN fold only
(already
204     enforced in experiment._calibrate_and_tune`); wire out_of_fold_predictions` once an
arbiter exists.
205
206     4.3 The golden corpus + active learning (the highest-leverage lever)
207     The single biggest quality lever is more labeled matches
([`NEXT_STEPS.md`](NEXT_STEPS.md)):
208     "corpus growth = highest leverage". With ~6 ? ~16 videos:
209     - Grow via the Roora vendoring flow
([`GOLDEN_SET_VENDORING.md`](GOLDEN_SET_VENDORING.md)),
210     [ROORA_LABELING_GUIDELINES.md`](ROORA_LABELING_GUIDELINES.md)) ? court calibration,
IAA, the
211     3 non-negotiables (ignore dead time, every rally, one-contact = one shot).
212     - Active learning: stop labeling videos at random. Score the *unlabeled* corpus and
label the
213     videos where the local segmenter is most *uncertain* / most *divergent from Gemini* /
most
214     *diverse* (new venue, fps, lighting). §5.3 surveys the methods; the local-vs-Gemini
agreement
215     signal (§4.4) is a ready proxy for "where do we disagree most ? label that."
216
217     4.4 Shadow evaluation: Gemini-agreement as a free, unlimited signal (NOT a gate, NOT
training data)
218     We have exactly one strong reference available on *unlabeled* footage at inference time:
Gemini.

```

219 - `experiment.compare\_unlabeled()` already scores agreement (agreed / a-only / b-only) without labels.

220 - Use it as a **shadow metric**: run any candidate segmenter on a large unlabeled pool, measure

221 agreement-with-Gemini, and watch it as a **trend** ? cheap, unlimited, catches regressions the 6

222 golden videos can't. **It is a shadow signal only.** Golden labels remain the gate; Gemini is not

223 ground truth (it has its own errors) and ? per the licensing guardrail (§8) ? **its outputs may**

224 never become training data for owned weights.

225

226 **### 4.5 One-command rally-segmentation eval (close gap #6)**

227 Wrap the above into a single reusable entrypoint (a future PR ? not built here): given a segmenter

228 variant + the golden manifest, emit the full §4.1 metric set + the honest stats block + a leaderboard

229 row, and (optionally) the §4.4 shadow agreement on an unlabeled pool. Every iteration in §6 is then

230 "run the entrypoint, read the honest delta, promote-or-not."

231

232 ---

233

234 **## 5. External research synthesis (cited, adversarially verified)**

235

236 > Produced by a deep-research run (6 search angles ? 27 sources ? 126 candidate claims ? 25

237 > 3-vote-adversarially-verified, 23 confirmed / 2 refuted). **Cite, don't over-claim.** Two claims

238 > were **killed** in verification ? recorded in §5.5 so we don't repeat them. **License caveat:** the

239 > code repos below are research releases; **independently confirm** each repo's actual license is

240 > MIT/Apache-2.0/BSD before vendoring any of it (this survey did not verify every license file).

241 > **Governance:** none of these techniques require training on paid-LLM outputs ? all are compatible

242 > with our no-Gemini-training-data guardrail (§8).

243

244 **### 5.1 Windowing & boundaries (TAL/TAS) ? the explicit-boundary machinery we lack**

245 The recurring lesson: modern action-localization models have an **explicit boundary mechanism**;

246 our heuristic has none (it merges on a velocity+gap rule). The transferable **idea** is "predict a

247 boundary and cut there," but the deep models are built for **hundreds+** of training videos, so at

248 n=6?16 they are **research bets**, not drop-ins.

249

250 - **ASRF ? Action Segment Refinement Framework** (Ishikawa et al., WACV 2021, [arXiv:2007.06866](https://arxiv.org/abs/2007.06866)).

251 A shared feature extractor feeds **two branches**: frame-wise classification + a **Boundary**

252 Regression Branch; predicted boundaries refine/split the segmentation (+13.7% edit, +16.1%

253 segmental F1 over MS-TCN-era SOTA on 50Salads/GTEA/Breakfast). **Why it fits:** the boundary head

254 is exactly the missing "cut here" mechanism. **Two caveats:** (1) ASRF's headline goal is to

255 **reduce** over-segmentation (merge spurious fragments) ? the opposite of our over-merge; only the

256 **split-at-a-detected-boundary** sub-mechanism transfers. (2) The claim that ASRF's boundary branch

257 is a **model-independent post-hoc splitter** you can bolt onto any dense signal was **REFUTED**

258 (§5.5) ? you must train the two-branch model, not graft the branch. **Effort/payoff:**

research bet.

259

260 - **TriDet** (Shi et al., CVPR 2023, [arXiv:2303.07347](https://arxiv.org/abs/2303.07347)). A

261 **Trident-head** models each boundary as a *relative probability distribution* over

262 local bins (boundary = expected value over neighboring instants) ? built for imprecise/ambiguous

263 boundaries. 69.3% avg mAP on THUMOS14 (> ActionFormer 66.8% at ~75% latency), **convolution-only**

264 (SGP), no self-attention (attractive on a single local GPU). Ablation shows the Trident-head's

265 gain concentrates at strict tIoU=0.7 ? it buys *boundary precision*. **Effort/payoff:**

research bet (trained on 100s of videos); the portable piece is the relative-boundary idea.

267

268 - **ActionFormer** (ECCV 2022, [arXiv:2202.07925](https://arxiv.org/abs/2202.07925)).

The reference

269 single-stage anchor-free template: multiscale pyramid + local self-attention; classify

270 every moment + regress its boundaries (71.0% mAP@0.5 THUMOS14). This is the family

TriDet/TemporalMaxer/

271 CausalTAD extend, and the natural target if we ever graduate from heuristic+LogReg.

Research bet.

272

273 - **TemporalMaxer** (CVPRW 2023, [arXiv:2303.09055](https://arxiv.org/abs/2303.09055)) ?

the most

274 important small-data signal in this dimension. It replaces ActionFormer's self-

275 attention block with a **parameter-free max-pool** and *beats* it (67.7 vs 66.8 avg mAP) at **~4x**

fewer params.

276 The ablation is the takeaway: a **learned 1-D conv block** is the **WORST** (59.4,

277 overfits redundant features); parameter-free pooling wins. **Why it fits:** direct evidence to **keep our**

scorer

278 parameter-light (the numpy LogisticRegression over a handful of features is the

right choice at

279 our scale, not a placeholder to rush past). **Effort/payoff:** a discipline (free) ?

adopt the

280 principle, not the model. (Caveat: the overfitting evidence is about feature

281 redundancy on a mid-size set, a directional inference to our small-n setting.)*

282

283 - **CausalTAD** (2024, [arXiv:2407.17792](https://arxiv.org/abs/2407.17792)) ?

steering-away note.

284 Causal-attention masking (attend only to past/present) won EPIC-Kitchens/Ego4D 2024

285 tracks, but it adds **no** boundary-split module. Useful later for *online/streaming* rally

286 detection; **NOT** the over-merge fix. Listed so we don't mistake causal attention for the remedy.

287

288 - **Metrics** (the actionable cross-cut): the field leads with **segmental Edit**

(Levenshtein) score

289 and F1@k, not frame accuracy (MoF) or plain tIoU ? see §5.4.

290

291 **### 5.2 Shuttle-trajectory ? rally signal processing ? published badminton systems**

already do what we lack

292 This is the highest-confidence, lowest-effort dimension: in-domain papers operationalize

293 the exact boundary heuristics we're missing, in spirit license-compatible (re-confirm before

294 vendoring code).

295 - **See et al. ? Service & End-of-Rally Detection in Badminton** (2024 Springer chapter

296 [10.1007/978-3-031-69073-0_15](https://link.springer.com/chapter/10.1007/978-3-031-69073-0_15);

297 2025 SN Computer Science follow-on
[10.1007/s42979-025-03935-0](https://link.springer.com/article/10.1007/s42979-025-03935-0)).

298 ***(a) Rally END = a sustained-inactivity rule:** the rally ends when **80% of**
consecutive frames

299 show no shuttle motion** (~**95% accuracy** over 187 rallies). ***(b) Rally START**
(service) = a

300 learned classifier** over court geometry + player poses (~26 features/frame × 12
frames ?

301 CNN/BiLSTM, ****88.6%** accuracy). ****Why it fits:** the inactivity rule is *precisely***
the**

302 end-of-rally cut our windowing lacks ? a few lines, directly breaks 65 s mega-windows.
****? Caveats:****

303 the rule is a ***termination***, not a ***within-rally split***; results rest on paywalled
abstracts +

304 small test sets (35?187 rallies); the serve classifier needs the (default-OFF) person
cue + court

305 polygons. ****Effort/payoff:** the inactivity rule = **low-effort high-confidence win**;**
serve

306 classifier = medium.

307

308 - ****Hsu, Yu & Cheng ? trajectory + swing fusion** (Sensors 2024, 24(13):4372,**
309 [10.3390/s24134372](https://www.mdpi.com/1424-8220/24/13/4372)). Fusing TrackNet
trajectory with a

310 YOLOv7 swing detector via a Shot Refinement Algorithm raises hit detection ****F1 72.3%**
? 90.5%**

311 (over 69 BWF matches / 1582 rallies). Hits are found by ****trajectory direction-**
reversal / peak

312 detection** (convert (x,y)?(frame,y) wave peaks) then reconciled with the swing cue.
****Why it fits:****

313 validates our ****late/score-level fusion**** approach in-domain, and direction-
reversal/peak is a cheap

314 ****per-window feature**** marking contact events (a "rally has hit cadence" signal ? the
discriminator

315 a "shuttle-moving" heuristic lacks). ****? Important caveat (the one 2-1 vote):** this**
paper does

316 ***within-rally stroke/hit* segmentation, **not** rally start/end ? "it splits merged**
rallies" is ***our**

317 **extrapolation***; frame it as "hit cadence plausibly helps locate rally boundaries," not
a proven

318 rally-splitter. ****Effort/payoff:** the peak/direction-reversal feature = low-effort**
win (feed the

319 scorer); full swing fusion = medium.

320

321 - ****What best separates "rally" from "shuttle merely moving"*** (synthesis): sustained-**
inactivity

322 termination (See), hit cadence / direction-reversals (Hsu), net-crossings ?2 (our
Gate-A,

323 [``rally_gate.py``](../backend/pipeline/detectors/rally_gate.py)), two-sided court
occupancy (our

324 person features). None alone is sufficient; the spine (§4) says ****feed them all to the**
learned

325 scorer**.

326

327 **### 5.3 Small-data strategies ? most quality per label**

328 - ****Timestamp / sparse supervision** (Li et al., CVPR 2021,**
[arXiv:2103.06669](https://arxiv.org/abs/2103.06669);

329 TAS survey Ding et al., [arXiv:2210.10352](https://arxiv.org/pdf/2210.10352)). Label
****one frame per**

330 **action** ? iteratively refine pseudo frame-labels ? *comparable* to full supervision**
(50Salads:

331 timestamp 75.6% acc / 60.1 F1@50 vs full 83.7% / 70.1). ****Why it fits:** sparse per-**
rally timestamps

332 cost far less than dense boundary labels ? relevant if we ever train a frame-wise
model.

333 ****? Caveats:** "comparable" hides a real **~8?10 pt residual gap**; annotation savings**

are modest

334 (~35%, annotators still watch the whole clip); benchmarks are cooking/egocentric ?
transfer to 1-D

335 shuttle trajectories is **unproven**. **Effort/payoff:** research bet to pilot.

336 - **Parameter-light scorer** ? see TemporalMaxer (§5.1): keep the numpy
LogisticRegression; resist a

337 learned temporal net until the corpus is much larger. **Low-effort win** (a
discipline).

338 - **OPEN** (the research could NOT verify these ? honest gap): **active learning**
(which video/segment
339 to label next via uncertainty/diversity/core-set) and the **teacher-student** /
distillation governance

340 boundary* for using Gemini as a weak/pseudo-labeler returned **no surviving verified**
claim*. Our

341 pragmatic, guardrail-safe stance (not from the literature ? from our own design): pick
the next

342 video to label by **maximal local-vs-Gemini disagreement** (the §4.4 shadow signal) +
scorer

343 uncertainty + **venue/fps diversity**; and treat **Gemini** strictly as an inference-
time

344 weak-reference / eval oracle ? never a training-data source for owned weights (§8).
These two

345 topics deserve their own grounded research pass before we lean on them.

346

347 **### 5.4 Evaluation rigor at small n ? lead with segmental metrics**

348 - **Frame accuracy (MoF) and plain tIoU are BLIND to over-segmentation** (TAS survey,
Ding et al.,
349 [arXiv:2210.10352](https://arxiv.org/pdf/2210.10352); segmental-F1 origin Lea et al.,
CVPR 2017).

350 "The score can be high even when segments are fragmented." The over-merge that bit us
is exactly

351 what tIoU-F1 hides (a giant window still **overlaps** real rallies). **Action:** lead
the leaderboard

352 with the **segmental Edit (Levenshtein) score** (penalizes spurious
ordering/fragmentation) **and**

353 $F1@{0.1,0.25,0.5}$, paired with our existing `segment_count_ratio` + the merge/split
breakdown.

354 **We already have** $F1@k$, $MAP@tIoU$, segment-count-ratio, bootstrap CIs, paired sign-
test, Platt/
355 isotonic + ECE/Brier ? the only **missing** primary metric is the **Edit score** (low-
effort add).

356 - **OPEN** (not covered by a verified claim): **IAA** / label-noise level in the golden
set, and which

357 over-segmentation-robust loss (ASRF boundary/smoothing, MS-TCN truncated-MSE) best
tolerates it at

358 small n. Measure IAA as the corpus grows (Roora has an A/A overlap plan); revisit
robust losses

359 only when a frame-wise model is on the table.

360

361 **### 5.5 Refuted claims (recorded so we do NOT repeat them)**

362 1. **ASRF's boundary branch is model-independent ? a post-hoc splitter applicable to**
any dense

363 segmentation output." **REFUTED (0-3)**. **Implication:** we cannot bolt ASRF's
boundary head onto

364 our trajectory signal as a drop-in; the two-branch model must be trained together. ?
port the

365 **concept**, don't expect a plug-in.

366 2. **"A GFL-inspired Boundary Distribution Regression head is the mechanism most**
relevant to splitting

367 merged windows." **REFUTED (0-3)** (and the cited source had a suspect identifier).
Do not adopt

368 this framing as established.

369

370 ---

371

```

372    ## 6. Prioritized roadmap (cheap ? durable)
373
374    Every item is its own measured PR: default-OFF, scored on the §4 metric set over the
golden set
375    with honest stats (CI + sign-test), re-confirmed on the served windowing (the
eval?serve gate),
376    logged in [QUALITY_ITERATIONS.md](QUALITY_ITERATIONS.md). Ordered by the research's
377    effort/confidence tiering. Each carries a falsifiable success criterion ? if it
isn't met, the
378    item does not promote (and that null is itself recorded).
379
380    ### Definition of "bridging the gap"
381    Local rally segmentation, measured honestly LOVO-by-match on the golden set,
approaches the
382    Gemini reference (today 0.93 single-court / 0.66 multi-court tIoU-F1) ? but tracked
primarily
383    through the over-segmentation-sensitive metrics (segmental Edit + F1@k + low
merge/split rate),
384    since those, not bare tIoU-F1, capture our actual failure. A secondary, free signal is
rising
385    agreement-with-Gemini on the unlabeled pool (§4.4, shadow only).
386
387    ### Tier 0 ? Make the failure measurable FIRST (do before any fix)
388    - E1 · Over-segmentation-aware leaderboard. Add the segmental Edit (Levenshtein)
score to
389    `segmentation_metrics.py`; surface Edit + F1@{0.1,0.25,0.5} + `segment_count_ratio` +
a
390    merge-rate / split-rate breakdown (promote the
`scratch/compare_local_vs_gemini.py` matching
391    into a reusable `backend/eval/` helper) in the default `experiment.compare()` report;
add the
392    one-command rally-segmentation eval endpoint (§4.5). Success: the current
heuristic's
393    over-merge shows up as a number (low Edit, high merge-rate) on the 6 golden videos,
and a
394    baseline row lands in the leaderboard. (Low effort, high confidence ? research-backed
§5.4.)
395
396    ### Tier 1 ? Low-effort, high-confidence wins
397    - W1 · Bounded windowing (the direct over-merge fix). Add a `max_window_duration`
cap and a
398    sustained-inactivity split to `trajectory_to_action_windows` ? port See et al.'s
rule (split /
399    end a window when ~80% of consecutive frames over a trailing window show no shuttle
motion).
400    Config-gated, default preserves current behavior. Success: segmental Edit + F1@k
improve vs
401    CONTROL on the golden set (?F1 CI excludes 0 and sign-test agrees) with no
served-windowing
402    regression; merge-rate drops materially. (Win ? §5.2 See et al., ~95% end-of-rally
acc.)
403    - W2 · Rally-state features into the learned scorer (no special-casing). Wire the
already-built,
404    default-OFF cues as scorer feature columns: net-crossings (Gate-A), two-sided
motion, and a
405    new cheap trajectory feature ? direction-reversal / peak count (Hsu) as a hit-
cadence proxy;
406    audio hit-rate when the hit detector lands. Re-measure with W1 in place and on the
served
407    windowing ? recall person was 0.021 and audio +0.079 but both CIs spanned 0 at
n=6 on the
408    pre-fix over-merged candidates; the picture may change once windows are bounded.
Success: the
409    ablation (`ablation.py`) shows a cue with ?F1 CI clearing 0 + sign-test, surviving
served re-check

```

```

410      ? promote that cue; others stay default-OFF + retained (no deletion on a small-n
null). *(win/medium.)*
411      - **W3 · Keep the scorer parameter-light (a discipline, documented not coded).** Per
TemporalMaxer's
412      ablation (learned temporal conv overfits; parameter-free wins), resist replacing the
numpy
413      LogisticRegression with a learned temporal net until the corpus is much larger. Record
this as a
414      standing decision. *(Win ? §5.1 TemporalMaxer.)*
415
416      ### Tier 2 ? Medium effort
417      - **M1 · Serve-start classifier ? rally-start boundary** (See et al. pose/court, 88.6%).
Needs the
418      person cue + a **court-mask source** (manual polygon vs auto-homography ? *owner-
gated* per
419      [ `MULTI_SIGNAL_FUSION_PLAN.md` ](MULTI_SIGNAL_FUSION_PLAN.md) §9.1). **Success:**
start-boundary
420      timing error drops on golden; promotes only if it lifts segmental F1@k without served
regression.
421      - **M2 · Per-venue threshold calibration** against golden labels (the heuristic's
footage-fragility ?
422      it scored 0.157 on multi-court). Tune `velocity_threshold` + `merge_gap` so local
segment counts
423      track golden counts per venue. **Success:** multi-court F1 rises toward the Gen-0
head's 0.561
424      without harming clean-court F1. *(Velocity is already fps-normalized ? calibrate per-
venue, not
425      per-fps.)*
426
427      ### Tier 3 ? Research bets (durable; corpus-gated)
428      - **R1 · Owned segmenter with an explicit boundary branch** (the ASRF/TriDet *idea* ? a
boundary head
429      that predicts cut points ? ported into our owned TAL head, NOT the full deep model,
and **trained**,
430      not grafted (§5.5 refutation)). Per
[ `OWNED_MODEL_IMPLEMENTATION_PLAN.md` ](OWNED_MODEL_IMPLEMENTATION_PLAN.md):
431      **TAL head first, VLM later.** **Gate:** the central unverified assumption is whether
boundary-head
432      gains hold at our scale on trajectory features ? so **pilot only when the corpus is
materially
433      larger (~?16?30 matches)**, and only after W1/W2 establish the heuristic ceiling.
**Success:**
434      beats the tuned heuristic + LogReg on held-out matches (segmental Edit + F1@k),
honestly LOVO.
435      - **R2 · Timestamp/sparse supervision** to get more labeled rallies per annotator-hour
(§5.3) ? pilot
436      if/when we train a frame-wise model; research bet (residual gap, unproven on
trajectories).
437
438      ### Throughout ? corpus growth + active learning + label quality
439      - **Grow the golden set 6 ? ~16** via Roora
([ `GOLDEN_SET_VENDORING.md` ](GOLDEN_SET_VENDORING.md)) ?
440      the single highest-leverage lever ([ `NEXT_STEPS.md` ](NEXT_STEPS.md)).
441      - **Active learning:** select the next videos to label by max **local-vs-Gemini
disagreement** (§4.4)
442      + scorer uncertainty + venue/fps diversity (our pragmatic stance; §5.3 flags the
formal methods as
443      an open research pass).
444      - **Measure IAA / label noise** as the corpus grows (Roora A/A overlap); revisit over-
segmentation-
445      robust losses only when a frame-wise model is on the table (§5.4 open question).
446
447      ### Dead-ends / do-NOT (so we don't relitigate)
448      - Don't treat **causal attention** (CausalTAD) as the over-merge fix ? it has no
boundary-split (§5.1).

```

```

449 - Don't expect **ASRF's boundary branch as a post-hoc drop-in** on our dense signal ?
refuted (§5.5).
450 - Don't adopt a **learned temporal net** at n=6?16 ? TemporalMaxer evidence says it
overfits (§5.1).
451 - Don't chase **bare tIoU-F1 / frame accuracy** ? blind to the over-merge (§5.4).
452 - Don't let **Gemini outputs become training data** for owned weights ? inference/shadow
only (§8).
453
454 ---
455
456 ## 7. Project: plan, tracking & definition of done
457
458 This roadmap is run as **one tracked project ? "Bridge the local?Gemini rally-detection
gap."** It
459 **completes when all committed tiers land AND a single quality number is attributed to
the effort**,
460 measured honestly and logged. This §7 is the living burn-down (update the status column
as PRs merge;
461 log each result in [QUALITY_ITERATIONS.md])(QUALITY_ITERATIONS.md), numbers in
462 eval_baselines/experiment_leaderboard.json).
463
464 ### 7.1 The number (what we attribute to this effort)
465 Measured **leave-one-video-out, grouped by match, on the golden set** (grown to ~16):
466 - **Headline:** multi-court rally-segmentation **tIoU-F1**, 'baseline ? final', plus
**gap-closure %**
467 = '(final ? baseline) / (Gemini ? baseline)'. Today: heuristic **0.157** ? Gemini
**0.66**
468 (Gen-0 head already **0.561**). **Success bar (proposed, owner-adjustable): close ?
50% of the
469 heuristic?Gemini multi-court gap** ? i.e. local multi-court F1 ? ~0.41, ideally
beating Gen-0's 0.561.
470 - **Primary lens (over-segmentation-sensitive, the actual failure):** segmental
**F1@0.25** +
471 **Edit-score** + **merge-rate** ? these must improve even if tIoU-F1 moves little.
472 - **Reported with honest stats** (bootstrap CI + paired sign-test); a result counts only
if it also
473 holds on the **served** windowing (eval?serve gate). The **baseline is frozen at E1**
before any fix.
474 - **Tracked headline = the n=6 golden *aggregate* tIoU-F1@0.5** (LOVO; the robust unit,
not a single
475 video) via 'python -m backend.eval.rally_seg_eval'. The 0.157/0.66 figures above are
per-venue
476 **references**; computing an exact "% gap-to-Gemini closed" needs Gemini measured on the
**same** golden
477 set + metric ? **TODO** (don't mix the per-venue refs with the aggregate).
478
479 **Progress (cumulative, golden n=6 aggregate tIoU-F1@0.5):**
480
481 | after | F1 | merge_rate | ? vs baseline |
482 |---|---|---|---|
483 | E1 baseline (current heuristic) | 0.300 | 0.523 | ? |
484 | + W1 bounded windowing (cap=12, **default-OFF**) | **0.439** | **0.038** | **+0.139**
(6/0 sign-test) |
485 | + W2 net-crossings gate (mc=1, **default-OFF**) | **0.457** | 0.038 | **+0.157
cumulative** (W2 step +0.019, 6/0) |
486 | + W2 multi-feature scorer (person/audio) | _next (GPU)_ | | |
487
488 ### 7.2 Work items (status checklist)
489 Status: ? not started · ? in progress · ? done. (Each row = its own default-OFF,
measured PR.)
490
491 | ID | Deliverable | Tier | Depends on | Acceptance / promote criterion | Status |
492 |---|---|---|---|---|---|
493 | **E1** | merge/split-rate metric + F1@k surfaced + one-command 'rally_seg_eval'
entrypoint; **froze baseline** (#185) | 0 | ? | over-merge shows up as a number on the 6 golden

```

videos | ? baseline F1 0.300, merge_rate 0.523 |

494 | ****W1**** | Bounded windowing ? `max\_window\_duration` cap + inactivity split (See et al.) (#187) | 1 | E1 | seg F1@k ? vs baseline (CI?0 ****and**** sign-test), no served regression; merge-rate ? | ? ****?F1 +0.139 (6/0), merge_rate 0.523?0.038**** (default-OFF) |

495 | ****W2**** | Rally-state features (net-crossings ? #188 + ****never-empty safeguard**** ?; two-sided motion, ****direction-reversal/peak****, audio = follow-on) | 1 | E1, W1 | ?1 cue ?F1 CI clears 0 + sign-test, survives served re-check ? promote; others retained default-OFF | ? ****net-crossings ?F1 +0.019 (6/0)****, gate now ****do-no-harm**** (never zeroes a video); multi-feature scorer next (GPU) |

496 | ****W3**** | Keep scorer parameter-light (standing decision; TemporalMaxer evidence) | 1 | ? | documented; no learned temporal net until corpus ? now | ? |

497 | ****M1**** | Serve-start classifier ? rally-start boundary (pose/court) | 2 | W1, ****court-mask decision (owner-gated)**** | start-boundary timing error ?; promote on seg-F1 lift, no served regression | ? |

498 | ****M2**** | Per-venue threshold calibration vs golden | 2 | E1 | multi-court F1 ? toward Gen-0 0.561 without harming clean-court | ? |

499 | ****R1**** | Owned segmenter with an explicit ****boundary head**** (port ASRF/TriDet ****idea****, trained ? not grafted) | 3 (stretch) | corpus ?~16?30, W1/W2 | beats tuned heuristic+LogReg on held-out matches (seg Edit + F1@k), LOVO | ? |

500 | ****R2**** | Timestamp/sparse-supervision pilot (label-efficiency) | 3 (stretch) | R1 path | measured label-efficiency win | ? |

501 | ****C**** | Golden corpus 6 ? ~16 (Roora) + active-learning selection + measure IAA | ongoing | ? | corpus ?16; IAA recorded | ? (~6 now; ~10 by the weekend) |

502

503 **### 7.3 Sequencing**

504 `E1` (metrics + baseline) ? `W1`/`W2`/`W3` (the high-confidence wins) ? `M1`/`M2` ? `R1`/`R2`

505 (corpus-gated). `C` (corpus growth) runs in parallel throughout ? it is the single highest-leverage

506 input and gates the Tier-3 bets.

507

508 **### 7.4 Definition of done**

509 The project is ****complete**** when:

510 1. ****Tier 0 + Tier 1 + Tier 2**** work items are merged ? each a measured PR (default-OFF, promoted only

511 on measured lift, re-confirmed on the served windowing).

512 2. ****The headline number (\$7.1) is recorded**** in `QUALITY\_ITERATIONS.md` + the leaderboard, with honest

513 stats, on the grown golden set ? ****and the gap-closure target is met, or the honest null is recorded**

514 with a reasoned go/no-go on Tier 3.**

515 3. Each landed item left the harness greener (no regression) and `QUALITY\_ITERATIONS.md` tells the story.

516

517 > ****Tier-3 scope decision (owner-adjustable):**** R1/R2 (the owned boundary-head model + sparse

518 supervision) are ****corpus-gated stretch goals**** ? the verified research says deep TAL detectors risk

519 > overfitting at n=6?16. ****Default:**** the project ****completes**** at Tier 0?2 + an attributed number, and

520 > Tier 3 spins out into the owned-model milestone ([`OWNED\_MODEL\_IMPLEMENTATION\_PLAN.md`](OWNED_MODEL_IMPLEMENTATION_PLAN.md)).

521 > Fold Tier 3 into the DoD instead if you want the project to span the owned model.

522

523 ---

524

525 **## 8. Constraints & guardrails (non-negotiable) + cross-links**

526

527 - ****Licensing:**** every dependency ? code, data, AND weights ? must be ****MIT / Apache-2.0 / BSD****.

528 Reject viral/NC/custom (Gemma-3, Llama-4 MAU cap, Commons-Clause).

529 - ****Never train owned weights on paid-LLM (Gemini/OpenAI/Anthropic) outputs**** (terms/copyright);

530 paid LLMs are inference/serving/shadow-reference only. Also forbidden: public datasets known to

```

531     be LLM-generated. **Exclude minors** from any training pool; per-user data needs
explicit opt-in.
532     - **Owned model:** TAL head FIRST (de-risked, trains in minutes, doubles as a pseudo-
label teacher +
533     honest baseline the VLM must beat), then Qwen3-VL (Apache-2.0) / InternVL3 (MIT) ? see
534     [ `OWNED_MODEL_IMPLEMENTATION_PLAN.md` ] (OWNED_MODEL_IMPLEMENTATION_PLAN.md).
535     - **Cross-links:** [ `HOW_RALLY_DETECTION_WORKS.md` ] (HOW_RALLY_DETECTION_WORKS.md) .
536     [ `EXPERIMENT_HARNESS.md` ] (EXPERIMENT_HARNESS.md) .
[ `MULTI_SIGNAL_FUSION_PLAN.md` ] (MULTI_SIGNAL_FUSION_PLAN.md) .
537     [ `ANALYZERS_AND_RECONCILERS.md` ] (ANALYZERS_AND_RECONCILERS.md) .
[ `ABLATION_LAYER.md` ] (ABLATION_LAYER.md) .
538     [ `EVALUATION.md` ] (EVALUATION.md) . [ `QUALITY_ITERATIONS.md` ] (QUALITY_ITERATIONS.md) .
539     [ `GOLDEN_SET_VENDORING.md` ] (GOLDEN_SET_VENDORING.md) .
[ `OWNED_MODEL_IMPLEMENTATION_PLAN.md` ] (OWNED_MODEL_IMPLEMENTATION_PLAN.md).
540

```

12. assistant (2026-06-16T15:15:01.058Z)

The cited file `output/local\_vs\_gemini\_6.json` doesn't exist. The numbers in the doc are *aggregate/per-venue* references, not per-video ground-truth scores for mahadevpura-2. Let me check the leaderboard and the QUALITY_ITERATIONS log for the actual recorded per-video numbers, especially the mahadevpura-2 hard-cap F1@0.25 0.722 / mIoU 0.50 and the count figures (n=40, count_ratio 1.03, mean 13.8s, max 43.3s).

13. user (2026-06-16T15:15:02.477Z)

```

1      {
2      "name": "gemini_wrapper_baseline",
3      "description": "Commodity-VLM wrapper baseline: the full pipeline's `gemini` segmenter
scored vs human golden labels at IoU 0.5. The commoditization/moat reference the owned model
must be judged against ? on multi-court footage the wrapper drops 0.93 -> 0.66, so the owned
model's target is the multi-court number, not the clean-footage one.",
4      "model": "gemini-flash-latest (resolved 2026-06-10; 3.5-flash era)",
5      "conditions": "serial uploads (ai_max_workers=1), chunks re-encoded to 1280px w/ audio
(ai_chunk_scale_width), 6-attempt exponential generate retry; min_segment_duration=3.0 filter
active (rejects real ~2.5s rallies ? see notes)",
6      "per_video": {
7      "mahadevpura_singles": {
8      "f1": 0.93, "precision": 1.00, "recall": 0.87, "mean_iou": 0.80,
9      "boundary_err_s": 0.7, "n_gt": 31, "n_pred": 27,
10     "notes": "3 of 4 FNs were detected by Gemini but rejected by
min_segment_duration=3.0 (2.5s rallies). Chunk 6 (unlabeled tail) lost to 503s ? no effect on
the labeled window."
11     },
12     "kushagra_singles": {
13     "f1": 0.66, "precision": 0.69, "recall": 0.62, "mean_iou": 0.73,
14     "boundary_err_s": 1.2, "n_gt": 32, "n_pred": 29,
15     "notes": "Multi-court (contrast 1.4x). Full 7/7 chunk coverage, audio intact. FPS
cluster as background-game pickups + one 53s dead-time merge ? the exact confound the contrast
metric predicts."
16     }
17     },
18     "comparison_at_time_of_run": {
19     "heuristic_lovo": {"mahadevpura_singles": 0.657, "kushagra_singles": 0.157},
20     "gen0_learned_lovo": {"mahadevpura_singles": 0.744, "kushagra_singles": 0.561}
21     },
22     "reliability_note": "gemini-flash-latest served 503 capacity waves throughout (one
chunk needed a ~5min backoff); a sole-provider dependency is operationally flaky ? the provider-
registry fallback design is load-bearing.",
23     "cost_note": "well under $1 for both videos at flash pricing (serial, ~14 chunk
requests total)."
24     }
25

```


14. user (2026-06-16T15:15:03.967Z)

Found 34 files

```
docs\RALLY_QUALITY_RESEARCH.md
docs\QUALITY_ITERATIONS.md
tests\test_rally_seg_eval.py
backend\eval\rally_seg_eval.py
tests\test_promotion.py
tests\test_segmentation_metrics.py
tests\test_served_gate_a_regression.py
tests\test_eval_metrics.py
tests\test_experiment_honest.py
tests\test_eval_batch.py
eval_baselines\fixtures\multi_rally_with_deadtime\expected.json
eval_baselines\fixtures\occlusion_break\expected.json
eval_baselines\gemini_wrapper_2026-06-10.json
eval_baselines\fixtures\clean_single_rally\expected.json
docs\archives\quality-iterations\2026-06-first-ab-experiment\README.md
docs\archives\quality-iterations\2026-06-human-gt-tuning\README.md
docs\archives\quality-iterations\2026-06-multivideo-lovo\README.md
docs\archives\research\OWNED_MODEL_TRAINING_STUDY.md
eval_baselines\experiment_leaderboard.json
docs\CODE_MAP.md
docs\ANALYZERS_AND_RECONCILERS.md
backend\eval\golden_fixtures.py
backend\eval\harness.py
backend\eval\metrics.py
backend\eval\segmentation_metrics.py
backend\eval\served_gate_a.py
backend\eval\ablation.py
backend\eval\batch.py
backend\eval\calibrate_wasb.py
backend\eval\calibration.py
backend\eval\eval_partial_labels.py
backend\eval\experiment.py
backend\eval\export_wasb_segments.py
backend\eval\fusion_compare.py
```

15. assistant (2026-06-16T15:15:08.858Z)

The Gemini wrapper baseline shows mahadevpura_singles with n_gt 31 (not 39 or 40) ? but that's a different clip/labeling than "mahadevpura-2". Let me search QUALITY_ITERATIONS.md and the leaderboard for the specific hard-cap numbers cited in the claim.

16. user (2026-06-16T15:15:11.007Z)

```
72-| tIoU-F1@0.5 | 0.439 | **0.457** | **+0.019** (95% CI [+0.011,+0.027], **sign-test 6/0**,
p=0.031) |
73-| F1@0.25 | 0.772 | 0.805 | +0.033 (6/0) |
74-| segment_count_ratio | 1.44 | 1.33 | over-split trimmed |
75-
76-Small but **statistically robust** (every video improves) and **served-measured** ? and a
satisfying reversal of the
--
97-|---|---|---|---|
98-| tIoU-F1@0.5 | 0.300 | **0.439** | **+0.139** (95% CI clears 0; **sign-test 6/0**, p=0.031)
|
99-| F1@0.25 | 0.501 | **0.772** | +0.271 |
100-| merge_rate | 0.523 | **0.038** | ?0.485 (over-merge essentially eliminated) |
101-| segment_count_ratio | 0.700 | 1.44 | now mildly over-*splits* ? W2's scorer to clean up |
102-
103-**Robust, not overfit:** every cap in 6?20 s clears 0 with 6/0 sign-test (the win isn't a
knife-edge); smaller caps trade
--
128-|---|---|
```

```

129-| tIoU-F1@0.5 | **0.300** [0.119, 0.494] |
130-| F1@0.25 | 0.501 [0.239, 0.747] |
131-| **merge_rate** | **0.523** [0.258, 0.788] |
132-| segment_count_ratio | 0.700 [0.405, 0.957] |
133-
134-The `0.300` **matches the independent served-side measurement below** (min_crossings=0 =
`wasb_hybrid` = 0.300) ? cross-validated.
--
323- match) so a *learned* producer's outputs used as arbiter features are out-of-fold (no
leakage).
324-- **C4 ? over-segmentation-visible metrics.** `backend/eval/segmentation_metrics.py`:
`f1_at_overlaps`
325: (F1@{10,25,50}), `average_precision`/`mean_average_precision` (mAP@tIoU),
`segment_count_ratio`. tIoU alone
326- is blind to a rally split in 3 or two rallies merged ? exactly what the reconciler exists
to fix.
327-

```

17. assistant (2026-06-16T15:15:15.794Z)

The QUALITY_ITERATIONS.md numbers are the n=6 golden *aggregate* (F1@0.5 0.300?0.439?0.457, F1@0.25 0.501?0.772?0.805). None of these are the per-video mahadevpura-2 hard-cap numbers (F1@0.25 0.722, mIoU 0.50) cited in the claim. Let me check the leaderboard JSON and search for the specific mahadevpura-2 per-video breakdown and the W1 count figures (n=40, count_ratio 1.03, mean 13.8s, max 43.3s).

18. user (2026-06-16T15:15:17.623Z)

```

107-      "owner_approved": true,
108-      "real_video_corroboration": "GX010125_1080p offline re-score: 45 candidates -> CONTROL
45 rallies vs Gate-A 27 (dropped 18 low-crossing held-shuttle/feed windows, ~1.7min)",
109:      "over_seg_segment_count_ratio": {"before": 1.68, "after": 1.01},
110-      "source_archive": "docs/archives/quality-iterations/2026-06-first-ab-
experiment/README.md"
111-    },

```

19. assistant (2026-06-16T15:15:21.253Z)

The leaderboard doesn't contain the mahadevpura-2 per-video numbers either. Let me do a broad search for where the specific cited figures (0.722, mIoU 0.50, count_ratio 1.03, mean 13.8, max 43.3, the 40 hand-labels) actually live ? possibly in the artifacts/ dir (untracked, just created) or a scratch report.

20. user (2026-06-16T15:15:22.890Z)

```

---OUTPUT DIR---
total 292
drwxr-xr-x 1 avidu 197609      0 Jun 16 20:34 .
drwxr-xr-x 1 avidu 197609      0 Jun 16 11:24 ..
drwxr-xr-x 1 avidu 197609      0 Jun 16 11:14 chunks
drwxr-xr-x 1 avidu 197609      0 Jun 16 20:38 highlights_work
-rw-r--r-- 1 avidu 197609 212992 Jun 16 20:34 val.db
-rw-r--r-- 1 avidu 197609 61195 Jun 16 20:38 val_extract.log
drwxr-xr-x 1 avidu 197609      0 Jun 16 20:38 val_reels
drwxr-xr-x 1 avidu 197609      0 Jun 16 20:34 wasb
---root json---
config.json

```

21. user (2026-06-16T15:15:23.775Z)

C:\Users\avidu\Projects\badminton-highlight-indexer\.claude\worktrees\w2-safeguard\docs\ANALYZERS_AND_RECONCILERS.md-584-****In-domain prior art that already hard-codes our cues**** (so our delta is the ***no-branch discipline***, not the docs\ANALYZERS_AND_RECONCILERS.md-585-cues): non-broadcast badminton rally start/end (See et al.

2024/2025), shuttle-state/hit detection (MonoTrack
docs\ANALYZERS_AND_RECONCILERS.md-586-arXiv:2204.01899; Hsu/Yu/Cheng, Sensors 2024 24(13):4372),
service-fault (Goh et al., Sensors 2023 23(24):9759),
docs\ANALYZERS_AND_RECONCILERS.md:587:tennis in-play state machines
(10.1080/19346182.2013.819007), badminton point-segmentation (Ghosh et al. 2018,
docs\ANALYZERS_AND_RECONCILERS.md-588-arXiv:1712.08714), ShuttleSet (arXiv:2306.04948).
C:\Users\avidu\Projects\badminton-highlight-
indexer\.claude\worktrees\w2-safeguard\docs\ANALYZERS_AND_RECONCILERS.md-589-
C:\Users\avidu\Projects\badminton-highlight-
indexer\.claude\worktrees\w2-safeguard\docs\ANALYZERS_AND_RECONCILERS.md-590-### 13.2
Terminology to adopt (gloss our nouns with the field's on first use)
--
C:\Users\avidu\Projects\badminton-highlight-
indexer\.claude\worktrees\w2-safeguard\docs\ANALYZERS_AND_RECONCILERS.md-657- arbiter can't
represent AND/XOR; hand-crossed products recover it without leaving "signal not verdict."
C:\Users\avidu\Projects\badminton-highlight-
indexer\.claude\worktrees\w2-safeguard\docs\ANALYZERS_AND_RECONCILERS.md-658- *(Snoek 2005.)* ?
producers + scorer.
C:\Users\avidu\Projects\badminton-highlight-
indexer\.claude\worktrees\w2-safeguard\docs\ANALYZERS_AND_RECONCILERS.md-659-- **C8 ? represent
a missing cue as an explicit `is\_present`/uncertainty column, not a silent 0** ? "cue
docs\ANALYZERS_AND_RECONCILERS.md:660: absent" ? "cue says no" for a linear arbiter. *(DBF
arXiv:1511.03183.)* ? **producers + scorer.**
C:\Users\avidu\Projects\badminton-highlight-
indexer\.claude\worktrees\w2-safeguard\docs\ANALYZERS_AND_RECONCILERS.md-661-- **C9 ? keep warm-
up/session-state a SOFT feature, never a hard pre-filter**, and add a court/context
docs\ANALYZERS_AND_RECONCILERS.md-662- invariance probe: the static court co-occurs with
rallies, so the scorer can learn "court visible" instead
C:\Users\avidu\Projects\badminton-highlight-
indexer\.claude\worktrees\w2-safeguard\docs\ANALYZERS_AND_RECONCILERS.md-663- of "point in
play" (action-context confusion). Evaluate on warm-up stretches that share the court but lack
--
docs\archives\quality-iterations\2026-06-first-ab-experiment\README.md-47-**Per-video held-out
F1 (IoU 0.5):**
C:\Users\avidu\Projects\badminton-highlight-
indexer\.claude\worktrees\w2-safeguard\docs\archives\quality-iterations\2026-06-first-ab-
experiment\README.md-48-| video | CONTROL | Gate-A | learned (LOVO) |
C:\Users\avidu\Projects\badminton-highlight-
indexer\.claude\worktrees\w2-safeguard\docs\archives\quality-iterations\2026-06-first-ab-
experiment\README.md-49-|---|---|---|---|
docs\archives\quality-iterations\2026-06-first-ab-experiment\README.md:50:| adarsh_avi_singles |
0.597 | **0.722** | 0.613 |
C:\Users\avidu\Projects\badminton-highlight-
indexer\.claude\worktrees\w2-safeguard\docs\archives\quality-iterations\2026-06-first-ab-
experiment\README.md-51-| kushagra_singles | 0.154 | **0.163** | 0.163 |
C:\Users\avidu\Projects\badminton-highlight-
indexer\.claude\worktrees\w2-safeguard\docs\archives\quality-iterations\2026-06-first-ab-
experiment\README.md-52-| largetest_doubles | 0.593 | **0.667** | 0.535 |
C:\Users\avidu\Projects\badminton-highlight-
indexer\.claude\worktrees\w2-safeguard\docs\archives\quality-iterations\2026-06-first-ab-
experiment\README.md-53-| gbaaddy | 0.231 | **0.274** | 0.222 |
--
C:\Users\avidu\Projects\badminton-highlight-indexer\.claude\worktrees\w2-
safeguard\docs\archives\research\OWNED_MODEL_TRAINING_STUDY.md-42-TAS head, given ASFormer's
small-data result used 2048-d I3D features?* This session answered it.
C:\Users\avidu\Projects\badminton-highlight-
indexer\.claude\worktrees\w2-safeguard\docs\archives\research\OWNED_MODEL_TRAINING_STUDY.md-43-
docs\archives\research\OWNED_MODEL_TRAINING_STUDY.md-44-- **Cross-video LOVO of the current
heuristic** (`calibrate\_local`, Adarsh-96 + Kushagra-32, human labels):
docs\archives\research\OWNED_MODEL_TRAINING_STUDY.md:45: mean held-out F1 **0.443 ± 0.279**;
Adarsh 0.722, **Kushagra 0.163** (its own ceiling ? *no* velocity-windowing
C:\Users\avidu\Projects\badminton-highlight-
indexer\.claude\worktrees\w2-safeguard\docs\archives\research\OWNED_MODEL_TRAINING_STUDY.md-46-
config does better). **The ±0.279 spread is method variance, not config.** ? This also
**corrects the study's

```
docs\archives\research\OWNED_MODEL_TRAINING_STUDY.md-47-  "~0.73 bar"**: that was a fit-on-self
single-match number; the honest cross-video heuristic bar is **~0.44**,
C:\Users\avidu\Projects\badminton-highlight-
indexer\.claude\worktrees\w2-safeguard\docs\archives\research\OWNED_MODEL_TRAINING_STUDY.md-48-
which *lowers* the Gen-0 ship gate.
--
C:\Users\avidu\Projects\badminton-highlight-
indexer\.claude\worktrees\w2-safeguard\docs\archives\quality-iterations\2026-06-multivideo-
lovo\README.md-34-```
docs\archives\quality-iterations\2026-06-multivideo-lovo\README.md-35-per-video best-fit:
adarsh 0.777 | kushagra 0.163 (ceiling ? NO config beats it)
docs\archives\quality-iterations\2026-06-multivideo-lovo\README.md-36-leave-one-video-out:
baseline mean 0.316 ? calibrated 0.443 ± 0.279 (overfit gap +0.027, tiny)
docs\archives\quality-iterations\2026-06-multivideo-lovo\README.md-37: held-out adarsh 0.722
(config tuned on Kushagra ? Adarsh's own optimum ? config TRANSFERS)
C:\Users\avidu\Projects\badminton-highlight-
indexer\.claude\worktrees\w2-safeguard\docs\archives\quality-iterations\2026-06-multivideo-
lovo\README.md-38- held-out kushagra 0.163 (config tuned on Adarsh = Kushagra's own ceiling)
C:\Users\avidu\Projects\badminton-highlight-
indexer\.claude\worktrees\w2-safeguard\docs\archives\quality-iterations\2026-06-multivideo-
lovo\README.md-39-```
C:\Users\avidu\Projects\badminton-highlight-
indexer\.claude\worktrees\w2-safeguard\docs\archives\quality-iterations\2026-06-multivideo-
lovo\README.md-40-**The asymmetry is the finding.** Tuning generalizes fine *where the heuristic
can solve the video*
docs\archives\quality-iterations\2026-06-multivideo-lovo\README.md-41:(Kushagra?Adarsh = 0.722).
But **Kushagra is unsegmentable by *any* velocity-windowing config (0.163)** ? and
docs\archives\quality-iterations\2026-06-multivideo-lovo\README.md-42-this is NOT a WASB-recall
failure: WASB tracks the shuttle **better** on Kushagra than Adarsh (**97% of GT
docs\archives\quality-iterations\2026-06-multivideo-lovo\README.md-43-rallies contain ?2 net-
crossings**, median 5, 310 visible pts/rally, vs Adarsh 76%). The likely mechanism:
C:\Users\avidu\Projects\badminton-highlight-
indexer\.claude\worktrees\w2-safeguard\docs\archives\quality-iterations\2026-06-multivideo-
lovo\README.md-44-Kushagra's shuttle is visible 67.5% of the time (vs Adarsh 40%) ? tracked
*through* inter-rally activity ? the
--
C:\Users\avidu\Projects\badminton-highlight-
indexer\.claude\worktrees\w2-safeguard\docs\archives\quality-iterations\2026-06-multivideo-
lovo\README.md-145-### Learned prototype LOVO (n=5)
C:\Users\avidu\Projects\badminton-highlight-
indexer\.claude\worktrees\w2-safeguard\docs\archives\quality-iterations\2026-06-multivideo-
lovo\README.md-146-| held-out | per-frame AUC | learned F1 (LOVO) | oracle F1 | heuristic best |
C:\Users\avidu\Projects\badminton-highlight-
indexer\.claude\worktrees\w2-safeguard\docs\archives\quality-iterations\2026-06-multivideo-
lovo\README.md-147-|---|---|---|---|---|
docs\archives\quality-iterations\2026-06-multivideo-lovo\README.md-148-| Adarsh (singles) |
0.879 | 0.709 | 0.722 | 0.73 |
C:\Users\avidu\Projects\badminton-highlight-
indexer\.claude\worktrees\w2-safeguard\docs\archives\quality-iterations\2026-06-multivideo-
lovo\README.md-149-| Kushagra (multi) | 0.853 | **0.561** | 0.633 | 0.16 |
C:\Users\avidu\Projects\badminton-highlight-
indexer\.claude\worktrees\w2-safeguard\docs\archives\quality-iterations\2026-06-multivideo-
lovo\README.md-150-| LargeTest (doubles) | 0.915 | 0.713 | 0.756 | 0.75 |
C:\Users\avidu\Projects\badminton-highlight-
indexer\.claude\worktrees\w2-safeguard\docs\archives\quality-iterations\2026-06-multivideo-
lovo\README.md-151-| gBaaddy (multi) | 0.833 | **0.574** | 0.574 | 0.28 |
```

22. assistant (2026-06-16T15:15:31.136Z)

This is an important finding. The cited per-video mahadevpura-2 numbers (F1@0.25 0.722, mIoU 0.50) do NOT appear anywhere in the tracked repo data. The only 0.722 figures that exist are for completely different videos (adarsh_avi_singles Gate-A 0.722; Adarsh held-out LOVO 0.722). The cited data file `output/local\_vs\_gemini\_6.json` does not exist. Let me check the val.db and the val_extract.log in output/ since those are the most recent (Jun 16 20:38) and might contain the actual June-12 validation run data referenced in the claim.

23. user (2026-06-16T15:15:33.293Z)

```
['videos', 'play_segments', 'sqlite_sequence', 'events', 'candidate_segments', 'jobs',  
'user_models']
```

24. user (2026-06-16T15:15:35.280Z)

No matches found

25. user (2026-06-16T15:15:40.114Z)

Traceback (most recent call last):

```
File "<string>", line 5, in <module>  
    for r in cur.execute('SELECT id, filename FROM videos').fetchall(): print(r)  
        ~~~~~~^AAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAA  
sqlite3.OperationalError: no such column: filename  
--- videos ---
```

26. user (2026-06-16T15:15:44.368Z)

```

videos cols: ['id', 'filepath', 'processed_at', 'status', 'validation_results', 'user_id']
('db84f3e65318e49aea8320de4c44f7d1', 'output\\highlights_work\\GX010128.MP4', '2026-06-16
12:32:39', 'processed', '[]', None)
('99a3e73afde3933e113767f2d66f71fe', 'output\\highlights_work\\GX010129.MP4', '2026-06-16
13:39:53', 'processed', '[]', None)
('6460c2fc7647e89537fc6afba80b519a', 'output\\highlights_work\\mahadevpura-0.MP4', '2026-06-16
14:20:31', 'processed', '[]', None)
--- play_segments by video ---
('6460c2fc7647e89537fc6afba80b519a', 62, 0.016683333333333335, 768.7513166666666)
('99a3e73afde3933e113767f2d66f71fe', 50, 0.016683333333333335, 536.7528833333333)
('db84f3e65318e49aea8320de4c44f7d1', 78, 0.016683333333333335, 755.70495)

```

27. assistant (2026-06-16T15:15:51.257Z)

This is very revealing. The val.db only contains 3 videos: GX010128 (78 segments), GX010129 (50), mahadevpura-0 (62). These match the "W1 cap=12" column counts in the prompt's table exactly (GX010128: 78, GX010129: 50, mahadevpura-0: 62). But mahadevpura-2 is NOT in this database at all.

So the actual local validation run that exists in the repo covered only 3 videos, and the W1-cap counts (78, 50, 62) correspond to those three. The mahadevpura-2 data (the entire ground-truth claim) is not present in any tracked or output file I can find. Let me confirm by searching the candidate_segments and the whole tree once more for the cited mahadevpura-2 ground-truth numbers (40 labels, mean 5.7s, F1@0.25 0.722, mIoU 0.50, boundary med 1.74/3.12).

28. user (2026-06-16T15:15:53.392Z)

Traceback (most recent call last):

```
File "<string>", line 7, in <module>
    for r in cur.execute('SELECT video_id, count(*) FROM events GROUP BY video_id').fetchall():
print(r)
```

```
sqlite3.OperationalError: no such column: video_id
candidate_segments by video:
('6460c2fc7647e89537ffc6afb80b519a', 62)
('99a3e73afde3933e113767f2d66f71fe', 50)
('db84f3e65318e49aea8320de4c44f7d1', 78)
events by video:
```

29. user (2026-06-16T15:15:54.225Z)

```
Found 6 files
docs\RALLY_QUALITY_RESEARCH.md
docs\archives\research\AUDIO_RALLY_DETECTION_PLAN.md
```

docs\archives\research\RALLY_DETECTION_RESEARCH.md
docs\PMF_MARKET_RESEARCH.md
docs\GOLDEN_SET_PHASE1_VIDEOS.md
docs\ANALYZERS_AND_RECONCILERS.md

30. assistant (2026-06-16T15:15:58.263Z)

Let me check these 6 files for any of the specific mahadevpura-2 ground-truth numbers, to be thorough before concluding the data isn't present.

31. user (2026-06-16T15:16:00.394Z)

110-- [Unstructured-video event detection ? arXiv 2002.08097](https://arxiv.org/pdf/2002.08097)
111-- [Badminton service-height CV setup (camera 4.5 m / 30°) ?
PMC](https://pmc.ncbi.nlm.nih.gov/articles/PMC10747833/)
112-- [VideoBadminton dataset ? arXiv 2403.12385](https://arxiv.org/html/2403.12385v1)
113-- [Table-tennis frame-rate vs detection accuracy ? Nature Sci
Rep](https://www.nature.com/articles/s41598-023-42966-6)
114-- [OpenTTGames (120 fps) dataset](https://lab.osai.ai/)

32. user (2026-06-16T15:16:02.476Z)

5ba6955 fix(fusion): W2 net-crossings gate is do-no-harm ? never empty a non-empty video (#188
follow-up)
64494ef feat(eval): golden regression fixtures M1 ? video-free CONTROL-only characterization
harness (#189)
a471745 feat(eval): Tier 1 (W2) ? net-crossings rally-state gate on W1 (?F1 +0.019, 6/0) (#188)
1de2817 docs: freeze Golden Regression Fixtures plan + adopt tracked-project-doc template (#186)
a3b209a feat(windowing): Tier 1 (W1) ? bounded windowing fixes the over-merge (?F1 +0.139, 6/0)
(#187)
90ab087 feat(eval): Tier 0 (E1) ? over-segmentation-aware rally-seg eval + frozen baseline
(#185)
960c00d docs: rally-quality research + eval-framework roadmap (foundation doc) (#184)
b6a39c4 feat(runner): fully-local no-AI segmenter mode (+ WASB stream flag, .env load) (#183)
---branch---
feat/w2-never-empty-safeguard
---untracked/modified---
M config.json

33. assistant (2026-06-16T15:16:14.446Z)

I now have a complete picture. Let me summarize what the data does and does not support.

The cited data file `output/local\_vs\_gemini\_6.json` does not exist anywhere in this worktree.
The only local validation database (`output/val.db`, the most recent run, Jun 16) contains
exactly 3 videos: GX010128, GX010129, and mahadevpura-0 ? and crucially, mahadevpura-2 is NOT
among them. The W1-cap counts in the val.db (78, 50, 62) match the prompt's "local W1 cap=12"
column for exactly those 3 videos.

Every specific numeric pillar of the mahadevpura-2 claim is absent from all tracked/output data
I can find:

- The hard-cap F1@0.25 0.722 / mean-IoU 0.50 / F1@0.5 0.296 figures appear nowhere. The only 0.722 values in the repo belong to entirely different videos (adarsh_avi_singles Gate-A 0.722; Adarsh LOVO held-out 0.722) ? a likely source of number-confusion.
- The W1 count figures for mahadevpura-2 (n=40, count_ratio 1.03, mean 13.8s, max 43.3s), the 40 owner hand-labels, mean 5.7s, and the boundary-error medians (1.74/3.12) do not appear in QUALITY_ITERATIONS.md, the leaderboard, the Gemini baseline, or val.db.
- The Gemini baseline that does exist (`gemini\_wrapper\_2026-06-10.json`) lists a "mahadevpura_singles" with n_gt 31, n_pred 27 ? different counts and a different clip than the "mahadevpura-2 / Gemini 39" cited.

Per the instruction "you MAY Read output/local_vs_gemini_6.json" and "ONLY the data," that data is not present, so the claim's evidence cannot be corroborated. I default to holds=false.

34. user (2026-06-16T15:16:26.090Z)

Structured output provided successfully

35. assistant (2026-06-16T15:16:32.187Z)

I have completed the adversarial verification and returned the result via StructuredOutput.

Summary of what I found: the claim does NOT hold against the available data. The cited evidence file `output/local\_vs\_gemini\_6.json` does not exist anywhere in the worktree, and every distinctive number in the mahadevpura-2 claim (the W1 counts, the 40-label ground truth, and especially the hard-cap F1@0.25 0.722 / mIoU 0.50) is absent from all tracked and output data. The only local validation database present (`output/val.db`) covers a different 3-video set that does not include mahadevpura-2, and the only 0.722 figures in the repo belong to unrelated videos (adarsh / Adarsh LOVO).